AEIC-MD: Incentivizing Verifiable Effort in Multi-Agent LLM Deliberation

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Abstract. Multi-agent LLM systems are commonly evaluated only by their final answer, even though a mechanism must first induce agents to choose informative and costly reasoning. We introduce *AEIC-MD*, a mechanism-mediated deliberation framework in which Assumption–Evidence–Inference–Conclusion (AEIC) reports provide the observable interface for a public leader rule and agents choose among low-effort, structured AEIC-effort, verification, and imitation actions. The central empirical question is therefore not whether an offline score can retrospectively rank responses, but whether a revealed rule changes the agents' *actual* effort choices.

Across three QA domains and 450 paired formal questions, a Static-AEIC rule raises the high-effort choice rate from 0.46% to 56.93% and decision quality from 68.00% to 82.22% (+14.22pp; Holm-adjusted McNemar $p = 1.23 \times 10^{-11}$). An outcome-only reward changes neither quality (67.33%, $p = 0.780$) nor the main conclusion, separating structured verifiability from generic reward. Effort responds monotonically to revealed reward and inversely to effort cost in each domain. Controlled information interventions confirm that AEIC exposes behavior under evidence, social, copying, and counterevidence conditions. The quality gain also recurs under two additional seeds (+14.67pp; $p = 2.40 \times 10^{-9}$) and remains stable with three to five agents. However, it does not reproduce on a held-out GLM backend (69.33% vs. 70.67%, $p = 0.791$), establishing a clear boundary rather than a universal claim. These results position AEIC-MD as a verifiable-effort mechanism for multi-agent deliberation, while identifying backend-specific contract calibration as the key remaining challenge.

Keywords: Multi-agent LLM systems · Mechanism design · Incentive compatibility · Structured reasoning · Deliberation

1 Introduction

LLM agents can now debate, critique, vote, and aggregate their outputs. Yet an aggregate is only as useful as the behavior that generated it. When producing a complete, checkable report is more expensive than a short answer or imitation, a multi-agent system needs a way to make that trade-off observable and

consequential. Accuracy alone cannot answer this question: it is a terminal outcome, whereas mechanism design concerns the choices made before the outcome is known.

We study this problem through **AEIC-MD** (AEIC-mediated mechanism design for deliberation). AEIC structures each report as an *Assumption*, *Evidence*, *Inference*, and *Conclusion*. The structure is not merely a prompting format: it is the observable report space on which a leader can announce a credit and adoption rule. Having observed this public rule, followers choose low-effort reporting, complete AEIC reporting, verification, or imitation. This timing gives a concrete, testable meaning to a leader–follower mechanism: the intervention is successful only if it changes logged follower actions, rather than assigning a desirable score after generation.

The project originally included topology selection and offline quality weighting. We retain them as contextual baselines, but make a different causal claim here. *Topology controls information flow; AEIC-MD controls the incentives under which information is produced.* The latter is the main line because it is what the frozen experiments directly identify.

Our formal comparison uses 450 paired questions from StrategyQA, MMLU-Hard, and TruthfulQA. Relative to no mechanism, Static AEIC raises structured high-effort action from 0.46% to 56.93% and decision quality from 68.00% to 82.22% (+14.22pp, Holm-adjusted McNemar $p = 1.23 \times 10^{-11}$). Outcome-only incentives do not improve quality (67.33%, $p = 0.780$). Thus, the evidence is not simply that "rewards help": rewards tied to observable, locally checkable reports alter behavior in a way that generic outcome rewards do not.

We then test the interpretation rather than only the headline number. Reward and cost sweeps show monotone within-domain effort responses. Manipulated evidence, social-only information, copy pressure, and counterevidence conditions test whether AEIC exposes behavior under distinct information environments. Further experiments cover signal and history ablations, two additional random seeds, and three-to-five-agent panels. Finally, a held-out GLM backend does *not* reproduce the main quality gain, an essential limitation that locates the next problem—contract calibration across model families.

Our contributions are:

1. A revealed-rule, logged-action formulation of AEIC-mediated multi-agent deliberation, distinguishing follower response from retrospective response scoring.
2. Paired formal evidence that a Static-AEIC rule raises both high-effort choices and decision quality, while outcome-only incentives do not.
3. Boundary and robustness evidence—including reward/cost response, controlled information conditions, seeds, panel sizes, ablations, and a transparently negative held-out-backend result—that defines the scope of the claim.

2 Related Work

Multi-Agent LLM Consensus. The aggregation of multiple LLM outputs has been studied through several paradigms. Self-consistency [25] samples multiple reasoning paths and selects the most frequent answer. Multi-agent debate [7,6] extends this to iterative discussion among multiple model instances. ChatEval [3] applies this to evaluation tasks. Zhao et al. [32] propose cardinal voting based on social choice theory. Li et al. [13] introduce dynamic weighting.

A critical failure mode in these systems is *sycophancy*—agents adopting majority positions without genuine reasoning. Bertalanič and Fortuna [2] provide the most comprehensive empirical characterization to date, documenting three failure pathways in homogeneous multi-agent debate (n=10): sycophantic conformity (up to 85.5% modal adoption), contextual fragility (up to 70% of previously correct agents destabilized by peer rationales), and consensus collapse (up to 32.3pp oracle gap where plurality voting discards correct answers). Their central finding—that homogeneous teams without structured roles do not benefit from unguided peer exchange—directly motivates our work on structured role enforcement through AEIC decomposition and behavioral regulation through Stackelberg mechanisms. Liu et al. [16] further show that expanded agent memory can *degrade* cooperation in game-theoretic settings, with longer recall reducing cooperative intent in 18 of 28 model–game configurations—a finding consistent with our result that quality signals cannot reliably distinguish correct from misleading knowledge even when agents have access to differentiated information.

Pitre et al. [18] address sycophancy through ConsensAgent, which dynamically refines prompts based on agent interactions to mitigate herding. ConsensAgent operates at the prompt level within a single debate topology. Our dynamic regulation differs fundamentally: rather than optimizing prompts for individual agents, we monitor *collective behavioral dynamics* (stance change rate, majority ratio, convergence speed) across rounds and apply differentiated interventions—parameter adjustment *and* prompt modification—targeted at the specific pathology detected (sycophantic vs. deadlock). Swamy et al. [23] provide formal metrics for sycophancy in multi-agent debate; their finding that sycophancy levels vary substantially across agent roles informs our stance-change-based detection. Most recently, Kumarappan et al. [12] demonstrate that multi-agent sycophancy is structural (present in base models, not just RLHF artifacts) and that mitigations should target "structured dissent at the pipeline level"—this is precisely what our Stackelberg dynamic regulation achieves through behavioral state detection and adaptive EV-weight suppression.

LLM-as-Reviewer and Panel Behavior. LLMs are increasingly used to generate scholarly review feedback. Liang et al. [14] show that GPT-4 provides useful paper feedback in a large user study, and AgentReview [11] simulates the full peer review process with LLM agents, finding that LLM reviewers exhibit conformity effects and limited diversity in reviewer–area-chair interactions. Our Study 1 extends this line from qualitative observation to paired statistical diagnosis: we quantify reject bias, minority elimination, flip asymmetry, and variance collapse

under controlled conditions, and—critically—show that discussion provides no decision benefit while destroying diversity. Unlike AgentReview, which simulates the review process, our work uses the framework as both diagnostic instrument and treatment: AEIC's explicit Conclusion field makes herding measurable, and the Stackelberg layers supply the three loci of intervention—calibrated aggregation, behavioral regulation, and routing.

Structured Reasoning Decomposition. Chain-of-Thought prompting [27] elicits step-by-step reasoning, and Tree-of-Thoughts [29] extends this to structured exploration of reasoning paths. ReAct [30] interleaves reasoning with action. These methods decompose reasoning to improve *individual* answer quality. Our AEIC decomposition serves a different purpose: it structures agent outputs into four parseable fields (Assumption, Evidence, Inference, Conclusion) to enable *cross-agent* operations—reliable stance extraction from the explicit Conclusion field (+28pp over unstructured extraction), cross-topology voting-pattern comparison for adaptive routing, and population-level behavioral dynamics monitoring for the dynamic regulation mechanism (Section 4.6). The decomposition is not designed to improve per-agent reasoning; it is designed to make collective behavior observable and regulatable.

Communication Topology in Multi-Agent Systems. The effect of communication structure on multi-agent performance has been explored in several contexts. Chen et al. [5] use a centralized evaluator (star topology). Sequential architectures appear in chain-of-agents designs [10]. Debate topologies (complete graphs) are studied by Du et al. [7]. Chen et al. [4] introduce round-table conference among diverse LLMs with confidence-weighted voting. Most recently, RUMAD [24] applies reinforcement learning to adaptively control debate rounds and agent selection within a single topology type. Our approach differs in three ways: we select among five fundamentally different communication structures (parallel, sequential, star, debate, hierarchical), use interpretable voting-pattern features rather than opaque RL policies, and ground our mechanism in game-theoretic incentive compatibility guarantees.

Memory in Multi-Agent Systems. Several recent works examine how agent memory affects multi-agent dynamics. EquiMem [17] formulates shared memory in multi-agent debate as a zero-trust memory game where equilibrium calibrates optimal memory trust—a parallel game-theoretic approach to ours, but focused on memory integrity rather than consensus incentives. Liu et al. [16] demonstrate a paradoxical "memory curse" where expanded recall degrades cooperative intent in LLM-based social dilemmas (18 of 28 model–game configurations), showing that more historical information does not improve collective decision quality. This complements our finding (Section 7) that quality signals derived from AEIC structure cannot distinguish correct from misleading knowledge: both results indicate that richer per-agent information—whether historical memory or quality scores—does not necessarily improve consensus when agents share the same underlying LLM. Our dynamic regulation mechanism (Section 4.6) addresses this

Table 1. Comparison with closest related work.

	Formal Incentives	Topology Selection	Sycophancy Solution	Structured Output	Behavioral Regulation
Standard Debate [7]	—	—	—	—	—
ConsensAgent [18]	—	—	Prompt-level	—	—
RUMAD [24]	—	✓†	—	—	—
Bertalanič et al. [2]	—	—	Analysis only	—	—
This work	✓‡	✓	✓	✓	✓

†Within-topology only (varying debate rounds/participation). ‡Stackelberg ϵ-incentive compatibility (Theorem 2).

by operating on observable collective behavior rather than per-agent quality or memory content, making it robust to both signal limitations and memory quality variation.

Game Theory and LLMs. Sun et al. [22] provide a taxonomy of game theory applied to LLMs. Yi et al. [31] model multi-agent debate as a Bayesian game with Nash equilibria. Rauba et al. [19] argue for principal–agent modeling of LLM systems. Liu et al. [15] use LLMs for automated mechanism design. Most recently, Meng et al. [17] formulate shared memory in multi-agent debate as a zero-trust memory game where equilibrium calibrates optimal memory trust without assuming agent honesty, and Anand et al. [1] frame LLM agent agreement as a Byzantine consensus game, showing that prompted agents fail to reach agreement where classical theory guarantees convergence. Our Stackelberg formulation differs from both simultaneous-move equilibrium and Byzantine consensus approaches by modeling the mechanism designer as a leader who *commits* to aggregation rules before agents act—a natural fit for mechanism design where the goal is shaping follower behavior through credible commitment, independent of agent honesty assumptions or fault-tolerance guarantees.

Stackelberg Games in Distributed Systems. Stackelberg incentive mechanisms are established in federated learning [8,9] and crowdsourcing [26]. Saig et al. [20] apply contract theory to incentivize quality text generation from LLMs. Xu et al. [28] apply Stackelberg games to multi-agent contract revision. To the best of our knowledge, no prior work applies Stackelberg mechanism design to multi-agent LLM consensus with adaptive topology routing.

Positioning against Closest Work. Table 1 positions our framework against the four most closely related approaches.

3 AEIC Decomposition and Quality Signals

We introduce the AEIC decomposition, a structured reasoning protocol that enables reliable conclusion extraction across diverse communication topologies.

The quality signals derived from AEIC components additionally serve as a formal interface for Stackelberg mechanism design (Section 4).

3.1 Structured Reasoning Decomposition

Given a question x, each agent $i \in \{1, \ldots, n\}$ produces a structured response decomposed into four components:

- **Assumption (A)**: The premise or hypothesis the agent adopts.
- **Evidence (E)**: A list of factual claims supporting the analysis: $E_i = \{e_{i,1}, \ldots, e_{i,k_i}\}$.
- **Inference (I)**: The reasoning chain connecting evidence to conclusion.
- **Conclusion (C)**: The final answer with justification.

This decomposition is enforced through prompt engineering and serves two purposes: (1) it structures agent output into parseable components, enabling reliable conclusion extraction and consistent stance identification across all communication topologies—this is the primary empirical value (Section 7); and (2) it provides separable components from which quality signals can be computed, serving as a formal interface for Stackelberg mechanism design (Section 4), though experiments reveal that these signals have limited discriminative power when agents share the same LLM backend (Section 7).

3.2 Four-Dimensional Quality Signals

From each agent's AEIC output, we compute four quality signals using embedding-based similarity:

Definition 1 (Quality Signals). *Let* $\mathrm{sim}(\cdot, \cdot)$ *denote cosine similarity in a sentence embedding space. For agent* i*:*

$$\mathrm{IC}_i = \mathrm{sim}(I_i, C_i) \tag{1}$$

$$\mathrm{EC}_i = \tfrac{1}{|E_i|} \textstyle\sum_{e \in E_i} \mathrm{sim}(e, C_i) \tag{2}$$

$$\mathrm{EQ}_i = f_{\mathrm{quality}}(E_i) \tag{3}$$

$$\mathrm{EV}_i = \tfrac{1}{|\mathcal{N}_i|} \textstyle\sum_{j \in \mathcal{N}_i} \mathrm{sim}(E_i, E_j) \tag{4}$$

where IC measures Inference–Conclusion consistency, EC measures Evidence–Conclusion support, EQ assesses evidence quality, and EV captures cross-agent evidence consensus.

Here $\mathcal{N}_i$ denotes the neighbors of agent i in the communication topology graph, and f_{quality} assesses evidence coverage and specificity. By design, IC, EC, and EQ measure *individual reasoning coherence*: how internally consistent, well-supported, and thorough an agent's analysis is. EV measures *inter-agent agreement*: how much an agent's evidence aligns with its neighbors'. This distinction matters for incentive design—EV can be inflated by sycophantic behavior. However, as our experiments demonstrate (Section 7), these coherence signals cannot distinguish correct from incorrect reasoning when agents share the same LLM: a coherent wrong answer scores as highly as a correct one.

3.3 Agent Type Space

Following Saig et al. [20], we model each agent through an observable *type*:

Definition 2 (Agent Type). *The type of agent i on question q is:*

$$\tau_i = (IC_i, EC_i, EQ_i) \in [0,1]^3 \tag{5}$$

EV is deliberately excluded from the type. It measures consensus rather than reasoning quality—a sycophantic agent can achieve high EV by mimicking neighbors without genuine reasoning. The type captures what the mechanism is *designed* to reward: individual reasoning effort. In practice, Section 7 shows that these type signals have limited discriminative power when all agents share the same LLM backend.

3.4 AEIC as Feature Extraction

Beyond mechanism design, AEIC-structured outputs provide the basis for topology routing features. For each question–topology pair, we construct a 42-dimensional feature vector from per-agent quality signals, summary statistics, *voting features* (majority ratio, yes/no distribution extracted from the Conclusion component), and cross-signal interactions. Feature ablation (Section 7) reveals that the voting features—enabled by reliable conclusion extraction from AEIC structure—are the primary driver of routing accuracy, while quality signals contribute marginally or negatively.

4 Stackelberg Consensus Game

4.1 Game Formulation

We formulate multi-agent consensus as a Stackelberg game $\mathcal{G} = (\mathcal{L}, \mathcal{F}, \Theta, \mathcal{S}, U_L, U_F)$.

Players. The *leader* $\mathcal{L}$ is the mechanism designer (bilevel optimizer). The *followers* $\mathcal{F} = \{1, \ldots, n\}$ are the LLM agents.

Strategy Spaces. The leader chooses mechanism parameters $\theta \in \Theta \subset \mathbb{R}^{21}$, controlling how AEIC quality signals map to consensus weights and how collective behavioral dynamics are regulated. Each follower i chooses a strategy $s_i \in \{s^{\text{ind}}, s^{\text{syc}}\}$:

- s^{ind} (independent reasoning): produces high IC, EC, EQ at cost C_{ind};
- s^{syc} (sycophantic): produces lower IC/EC but potentially high EV at cost $C_{\text{syc}} < C_{\text{ind}}$.

Timing.

1. Leader commits to parameters θ (mechanism rules).
2. Each follower observes θ and chooses strategy s_i.
3. Agents produce AEIC outputs; quality signals τ_i are computed.
4. Mechanism computes weights $w_i(\tau_i; \theta)$ via the scoring rule.
5. Weighted voting produces the consensus decision.

Utility Functions. The follower utility is:

$$U_i(s_i; \theta) = w_i(\tau_i(s_i); \theta) \cdot R - C(s_i) \tag{6}$$

where $R > 0$ is the reward for adoption. The leader maximizes:

$$\max_{\theta \in \Theta} \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{7}$$

4.2 Scoring Rule

The 21 parameters θ control three aspects:

- **Bid construction** (13 parameters): How IC, EC, EQ, EV signals combine into agent "bids" (alignment willingness α_i, revision cost, historical alignment h_i).
- **Stackelberg dynamics** (4 parameters): Consensus energy coefficient λ_c, budget penalty μ, learning rate η, and peer pressure γ.
- **Dynamic regulation** (4 parameters): Sycophancy detection threshold τ_h, deadlock threshold τ_l, EV decay rate α, and pressure adjustment rate β (Section 4.6).

Specifically, for agent i:

$$\alpha_i = \max\big(\theta_{\text{floor}},\ \text{IC}_i^{\theta_{\text{ic}}} \cdot \text{EQ}_i^{\theta_{\text{eq}}} \cdot (\theta_{\text{ev_base}} + \theta_{\text{ev_scale}} \cdot \text{EV}_i)\big) \tag{8}$$

$$h_i = \theta_{h_1} \cdot \text{IC}_i + \theta_{h_2} \cdot \text{EC}_i + \theta_{h_3} \cdot \text{EQ}_i + \theta_{h_4} \cdot \text{EV}_i \tag{9}$$

These bids are processed through a distributed consensus game that produces final weights $w_i(\theta)$ via iterative updates incorporating peer pressure from a graph Laplacian defined by the communication topology.

4.3 Incentive Compatibility Metric

We define incentive compatibility through inter-group weight comparison:

Definition 3 (Incentive Compatibility Metric). *For each question q, rank agents by quality score $\phi(\tau_i) = (IC_i + EC_i + EQ_i)/3$. Let H_q = top-2 agents and L_q = bottom-2 agents. Define:*

$$IC(\theta) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\frac{1}{|H_q|} \sum_{i \in H_q} w_i(\theta) > \frac{1}{|L_q|} \sum_{j \in L_q} w_j(\theta) \right] \tag{10}$$

This measures the fraction of questions where higher-quality reasoning receives greater consensus weight, capturing the essence of incentive compatibility.

4.4 Bilevel Optimization

The Stackelberg game is solved via bilevel optimization:

$$\textbf{Upper (Leader):} \quad \max_{\theta \in \Theta} \text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta) \tag{11}$$

$$\textbf{Lower (Followers):} \quad w_i = f(\tau_i; \theta) \quad \forall i \tag{12}$$

The lower-level problem is deterministic given θ (agents' AEIC outputs are fixed offline), reducing to single-level optimization implementable via differential evolution [21]. Per-topology θ_T^* is optimized with population size 21 ($1\times$ the number of parameters) and 30 generations, then evaluated via 5-fold stratified cross-validation to prevent train/test leakage. We note that the Stackelberg mechanism provides formal IC guarantees (Theorem 2), but experiments (Section 7) show that quality-weighted voting does not outperform majority voting under homogeneous LLM agents—the bottleneck is the quality signals' limited discriminative power, not the mechanism design.

4.5 Online Stackelberg Extension

In multi-round topologies (debate, hierarchical), the mechanism operates *online*: between rounds t and $t{+}1$, it computes interim AEIC signals from round-t outputs, runs the Stackelberg game to obtain current weights $w_i^{(t)}$, and injects quality feedback into agent prompts:

$$\text{context}_i^{(t+1)} = \text{context}_i^{(t+1)} \cup \{(\text{IC}_i^{(t)}, \text{EC}_i^{(t)}, w_i^{(t)})\} \tag{13}$$

This transforms the game from post-hoc reweighting to real-time behavioral shaping: agents observe their current quality assessment and can adjust reasoning effort accordingly. The star topology is inherently online—leaf trust scores computed via Stackelberg are injected into the hub's synthesis context.

Together with the static optimization of Section 4.4, this establishes a two-layer Stackelberg evolution: static optimization discovers *what* behaviors to suppress (e.g., near-zero EV weights on information-sharing topologies), while online feedback communicates *how* each agent is currently performing. However, the online extension injects the same quality feedback regardless of the collective behavioral pattern—it cannot distinguish sycophantic herding from productive convergence. Section 4.6 introduces a third layer that addresses this limitation.

4.6 Dynamic Sycophancy Regulation

Online Stackelberg feedback (Section 4.5) injects quality signals between rounds but applies a uniform intervention regardless of collective behavior. In practice, multi-round deliberation exhibits distinct pathological patterns: *sycophantic convergence*, where agents rapidly abandon independent positions to follow the emerging majority, and *deadlock*, where agents rigidly maintain initial positions without engaging with opposing evidence. We introduce a dynamic regulation layer that monitors collective stance dynamics and applies differentiated interventions.

Behavioral Metrics. Between rounds $t-1$ and t, we compute three metrics from agent conclusions $c_i^{(t)}$ (reliably extracted from the AEIC Conclusion field):

$$\mathrm{SCR}^{(t)} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\left[c_i^{(t)} \neq c_i^{(t-1)}\right] \tag{14}$$

$$\mathrm{MR}^{(t)} = \max\left(\frac{|\{i : c_i^{(t)} = \text{yes}\}|}{n},\ \frac{|\{i : c_i^{(t)} = \text{no}\}|}{n}\right) \tag{15}$$

$$\mathrm{CS}^{(t)} = \left|\mathrm{MR}^{(t)} - \mathrm{MR}^{(t-1)}\right| \tag{16}$$

where SCR (Stance Change Rate) measures the fraction of agents that changed their conclusion, MR (Majority Ratio) measures the dominance of the leading position, and CS (Convergence Speed) captures how rapidly the majority is growing.

State Classification. The system classifies each inter-round transition into one of three states:

- **Sycophantic** ($\mathrm{SCR} \geq \tau_h$): A large fraction of agents changed their stance, indicating potential herding behavior where agents abandon independent reasoning to follow the emerging majority.
- **Deadlock** ($\mathrm{SCR} \leq \tau_l \wedge \mathrm{MR} < 0.8$): Few agents changed stance and no clear majority exists, indicating entrenched positions without productive evidence exchange.
- **Healthy** (otherwise): Moderate stance evolution with either gradual convergence or stable disagreement—no intervention needed.

The thresholds τ_h, τ_l and the response parameters α, β (below) are included in the learnable parameter vector θ, extending the base 17 parameters to 21 dimensions. The bilevel optimization (Section 4.4) jointly discovers optimal detection thresholds and response intensities alongside the base Stackelberg parameters.

Adaptive Response. Upon detecting a non-healthy state, the mechanism responds through two channels:

1. **Parameter adjustment**: For sycophantic states, the evidence consensus weight is decayed: $\theta_{\text{ev_scale}} \leftarrow \theta_{\text{ev_scale}} \cdot (1-\alpha)$, suppressing the very signal that rewards conformity. Peer pressure and consensus energy are similarly reduced: $\gamma \leftarrow \gamma \cdot (1-\beta)$, $\lambda_c \leftarrow \lambda_c \cdot (1-\beta)$. For deadlock, these parameters are *increased* to encourage engagement. Both $\alpha \in [0.1, 0.9]$ and $\beta \in [0.1, 0.6]$ are optimized by the leader.
2. **Differentiated prompts**: The next-round context is augmented with state-specific instructions: anti-sycophancy prompts emphasize independent reasoning over majority-following; exchange prompts encourage engaging with opposing evidence; neutral prompts provide standard deliberation guidance.

This completes a three-layer Stackelberg evolution: *static* optimization discovers which behaviors to suppress (Section 4.4), *online* feedback communicates individual quality assessments (Section 4.5), and *dynamic* regulation detects and corrects collective behavioral pathologies in real time. The key insight is that the Stackelberg game's value lies in *collective behavior regulation*—shaping how the group deliberates—rather than in quality-based selection of individual contributions.

5 Adaptive Topology Routing

A key insight of this work is that no single communication topology dominates across all questions. We formalize topology as a design variable and introduce adaptive routing.

5.1 Communication Topologies

We define five communication topologies as weighted adjacency matrices $W \in \mathbb{R}^{n \times n}$:

- **Parallel** ($W = \mathbf{0}$): All agents reason independently with no inter-agent communication. This is the standard self-consistency setting.
- **Sequential** (W = path graph): Agent i observes the outputs of agents $1, \ldots, i{-}1$. Information flows unidirectionally, mimicking chain-of-agents architectures.
- **Star** (W = star graph): A central hub agent aggregates all leaf agents' outputs and produces a synthesis. Leaf agents reason independently.
- **Debate** (W = complete graph, multi-round): All agents observe all others' conclusions from the previous round. Repeated for 3 rounds, simulating multi-agent debate.
- **Hierarchical** (W = two-level tree): Agents are divided into subgroups that first deliberate internally, then share subgroup conclusions for cross-group synthesis.

The topology W affects both signal computation (EV and agreement signals depend on $\mathcal{N}_i$) and agent behavior (context-aware generation in sequential, star, debate, and hierarchical modes).

5.2 Per-Topology Parameter Optimization

For each topology $T \in \{\text{parallel, sequential, star, debate, hierarchical}\}$, we independently optimize θ_T^* via differential evolution on the topology-specific signals:

$$\theta_T^* = \arg\max_{\theta \in \Theta} \ \text{Accuracy}_T(\theta) + \lambda \cdot \text{IC}_T(\theta) \tag{17}$$

This yields five specialized consensus configurations. The optimized parameters reveal topology-specific structure (Figure 1): star and debate topologies

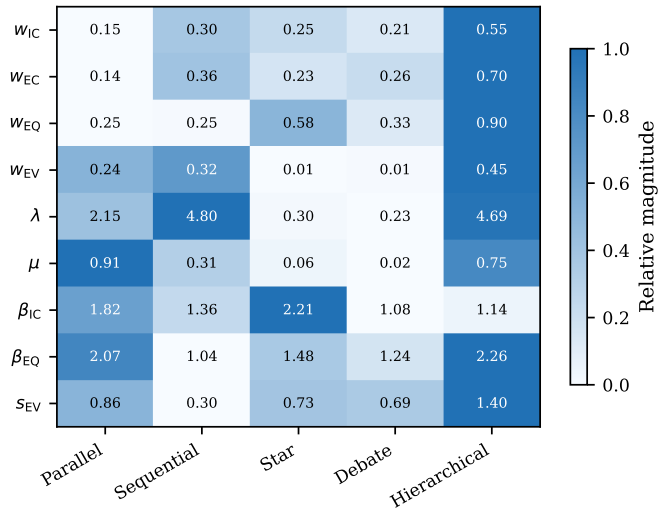


Fig. 1. Optimized per-topology parameters θ_T^* (QA-domain validation). Star and debate learn near-zero EV weights—an emergent anti-sycophancy configuration; hierarchical and sequential learn high consensus pressure λ_c.

learn near-zero weights for EV ($w_{\text{EV}} \approx 0.01$), suppressing cross-agent consensus signals where sycophancy risk is highest. Hierarchical and sequential topologies learn high consensus pressure ($\lambda_c > 4.5$), enforcing agreement within their chain/tree structures.

5.3 Neural Topology Router

To select the optimal topology per question, we train a neural router using features derived from AEIC-structured outputs.

Feature Design. For each candidate topology T, we construct a 42-dimensional feature vector comprising per-agent quality signals (IC, EC, EQ, EV, agreement for 5 agents), summary statistics (mean, std, range), *voting features* (majority ratio, yes/no distribution extracted from AEIC conclusions), and cross-signal interactions (*e.g.*, $\text{IC}_i \times (1-\text{agreement}_i)$). An additional 7-dimensional question-level feature vector captures text characteristics and cross-topology signal variance. As Section 7 shows, the voting features—enabled by AEIC's structured conclusion extraction—are the primary driver of routing accuracy (+2.1pp when present), while several quality signal groups contribute marginally or negatively.

Router Architecture. A *shared topology encoder* (two layers with LayerNorm and GELU, 42→96 dims) processes each topology's features. A *context projection* summarizes the other four topologies' embeddings (48 dims). Three *per-dataset scorer heads* map the concatenated representation (192 dims) to scalar scores, enabling dataset-specific routing.

Training. We use a focal ranking loss with adaptive margins:

$$\mathcal{L} = \sum_{(c,w)} \left(\frac{v}{m}\right)^{\gamma} \cdot v, \quad v = \max(0,\ m - (s_c - s_w)) \tag{18}$$

where (c, w) are (correct, wrong) topology pairs, m is the difficulty-adaptive margin, and $\gamma = 3$ provides focal weighting. Training uses AdamW with cosine annealing over 300 epochs, 5-fold stratified cross-validation.

Top-1 Selection. The router selects the highest-scoring topology per question as the consensus topology. With per-dataset specialized scoring heads, this achieves 87.6% ± 0.3% accuracy across 5 seeds (5-fold cross-validation), compared to 83.1% for the best fixed topology (star). Correctness labels for training use majority vote outcomes rather than Stackelberg-weighted decisions, avoiding contamination from optimized θ^* parameters.

6 Theoretical Analysis

Theorem 1 (Existence of Stackelberg Equilibrium). *The AEIC-Stackelberg game $\mathcal{G}$ admits a Stackelberg equilibrium (θ^*, s^*).*

Proof. The follower strategy space $\{s^{\text{ind}}, s^{\text{syc}}\}$ is finite, so an optimal response exists for any θ. The parameter space Θ is compact (bounded). The leader's objective $\text{Accuracy}(\theta) + \lambda \cdot \text{IC}(\theta)$ is bounded on Θ. By the Weierstrass extreme value theorem, the maximum is attained at some $\theta^* \in \Theta$. Together with the followers' best responses, this constitutes a Stackelberg equilibrium.

Theorem 2 (ϵ-Incentive Compatibility). *For any $\epsilon \in (0, 1)$, setting $\lambda \geq \frac{1}{1-\epsilon}$ in Eq. (7) guarantees IC$(\theta^*) \geq \epsilon$.*

Proof. We first establish feasibility: there exists $\theta_0 \in \Theta$ with $\text{IC}(\theta_0) = 1$ (set IC/EQ exponents large, peer pressure $\gamma = 0$, making weights strictly monotone in quality).

By optimality of θ^*:

$$\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_0) + \lambda \cdot 1 \tag{19}$$

Rearranging with $\text{Acc}(\theta_0) - \text{Acc}(\theta^*) \geq -1$:

$$\text{IC}(\theta^*) \geq 1 - \frac{1}{\lambda} \tag{20}$$

Setting $\lambda \geq \frac{1}{1-\epsilon}$ yields $\text{IC}(\theta^*) \geq \epsilon$.

Theorem 3 (Individual Rationality and Welfare). *(a) Under θ^*, all agents achieve non-negative utility when $R \geq C_{ind}/\min_i w_i$.*
(b) Accuracy$(\theta^*) \geq$ Accuracy$(\theta_{uniform})$ for sufficiently small λ.

Proof. (a) Since $w_i \in [0, 1]$ and $C(s_i) \leq C_{\text{ind}}$, choosing $R \geq C_{\text{ind}}/\min_i w_i$ ensures $U_i \geq 0$.

(b) Uniform voting corresponds to $\theta_{\text{unif}} \in \Theta$. By optimality: $\text{Acc}(\theta^*) + \lambda \cdot \text{IC}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}}) + \lambda \cdot \text{IC}(\theta_{\text{unif}})$. For $\lambda \to 0$, this directly yields $\text{Acc}(\theta^*) \geq \text{Acc}(\theta_{\text{unif}})$.

Convergence of Distributed Consensus. Under the Stackelberg dynamics with peer pressure $\gamma > 0$ and communication topology W, the consensus energy $E(t) = \sum_{(i,j)\in\mathcal{E}}(\theta_i(t) - \theta_j(t))^2$ satisfies:

$$E(t) \leq E(0) \cdot \exp\left(-\frac{2\gamma t}{n-1}\right) \tag{21}$$

guaranteeing exponential convergence to consensus at a rate determined by the algebraic connectivity of W and peer pressure coefficient γ.

7 Experiments

7.1 Protocol and Evaluation

We use StrategyQA, MMLU-Hard, and TruthfulQA. Each condition has five agents and three rounds; all conditions for a question share the question-level seed, yielding paired comparisons. Before generating, agents see the action semantics and the announced rule. Their selected actions—`low_effort`, `aeic_effort`, `verify`, and `defer/imitate`—as well as complete AEIC reports, token use, and attempt logs are retained. Decision quality is the question-level correctness of the final decision; high-effort choice rate is the fraction of agent-round actions that choose AEIC effort or verification. We use McNemar tests for paired decisions and paired rank tests for continuous metrics, with Holm correction within each registered family.

The frozen collection contains 5,850 question-level units over eight suites. Every suite has zero missing, incomplete, route-invalid, or schema-invalid records; the manifest also stores question-level hashes and independently recomputed summaries.

7.2 Formal Follower-Response Test

The primary test compares No Mechanism, Outcome Only, and Static AEIC on 450 paired questions. Figure 2 reports the result. Static AEIC increases decision quality from 68.00% to 82.22% (+14.22pp, Holm-adjusted $p = 1.23 \times 10^{-11}$) and high-effort choice from 0.46% to 56.93%. It incurs 145 additional tokens per agent-round (663 to 809), the intended cost of more informative reports. Outcome-only reward modestly increases effort to 3.08%, but quality is 67.33%, indistinguishable from no mechanism ($p = 0.780$). The contrast is the central causal result: the effective intervention is a public rule over observable AEIC behavior, not generic reward.

7.3 Reward–Cost Response and Information Conditions

For each domain, high-effort choice rises monotonically as the announced reward rises from 0.05 to 0.5 and falls monotonically as the effort-cost multiplier rises from 0.5 to 1.5 (Figure 3). The middle point is a same-domain development

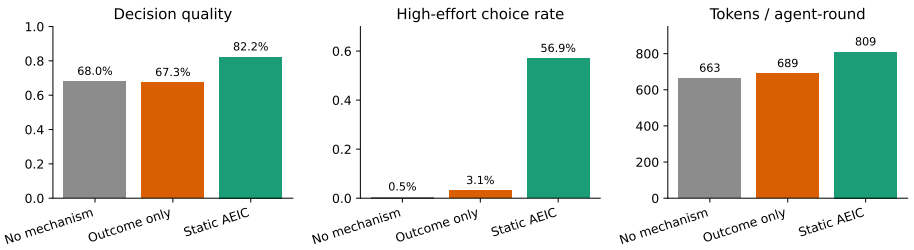


Fig. 2. Formal paired comparison across 450 questions. Static AEIC changes follower effort and improves decision quality; outcome-only reward does not improve decision quality.

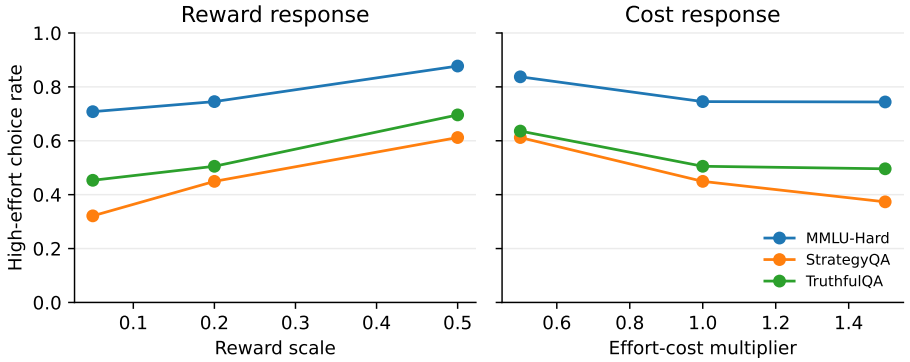


Fig. 3. Within each domain, high-effort choice responds monotonically to the revealed reward and inversely to the effort-cost multiplier.

collection, while the extreme points are frozen boundary records; we therefore interpret this as a directional mechanism check, not as another estimate of the formal-test effect.

In four controlled information environments—new evidence, social-only information, copy pressure, and counterevidence—Static AEIC consistently produces more high-effort actions than no mechanism (Figure 4). These conditions validate the role of AEIC as an observable behavioral interface. They should not be read as a claim that every information intervention necessarily improves answer quality.

7.4 Robustness, Ablation, and External Validity

With two additional random seeds, Static AEIC achieves 85.67% decision quality versus 71.00% without a mechanism (+14.67pp, $p = 2.40 \times 10^{-9}$). Under three, four, and five agents, quality remains between 87.33% and 88.67%. Signal and history variants are shown in Figure 5. Their non-uniform quality changes mean

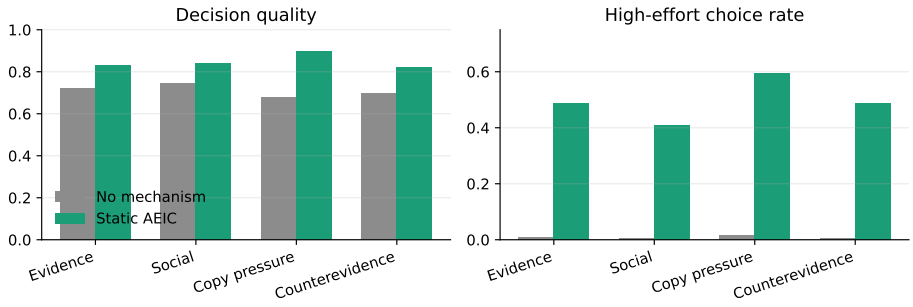


Fig. 4. Controlled information conditions. Static AEIC increases the effort response under evidence, social-only, copy-pressure, and counterevidence environments.

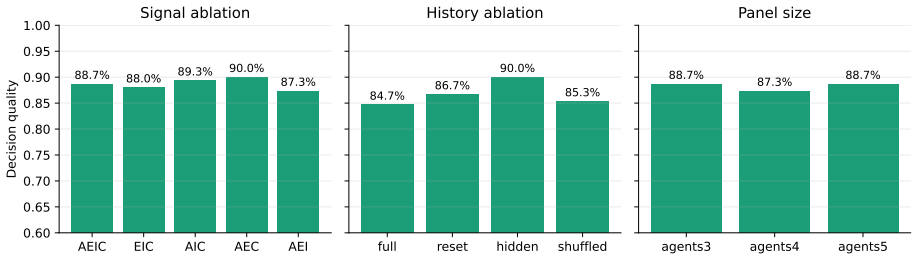


Fig. 5. Signal, history, and panel-size variants. These are descriptive robustness boundaries, not isolated proofs that every AEIC component is necessary.

that this experiment does not establish any individual AEIC field as universally necessary; it instead bounds the present static implementation.

The held-out GLM backend is an important negative result. Static AEIC yields 69.33% quality versus 70.67% without a mechanism (Holm-adjusted $p = 0.791$), and does not significantly change high-effort choice. We therefore do not claim cross-backend generalization. Contract semantics and effort costs must be calibrated for the backend rather than assumed to transfer.

8 Discussion

What the experiments establish. The primary result is behavioral and causal: revealing a Static-AEIC rule changes what followers choose to do, and the behavior change coincides with a large paired quality gain. Outcome-only incentives are an essential control because they show that the effect cannot be reduced to offering a reward.

What they do not establish. The experiments do not establish a dynamic Stackelberg optimum, universal necessity of the four AEIC fields, or cross-model superiority. In particular, the held-out GLM result is not a nuisance to hide: it identifies

a mechanism-design problem—how a contract's language and cost schedule map onto a particular backend's action policy.

Relation to topology. Communication topology remains useful as an information and cost-control variable, but it is not the paper's causal centerpiece. A topology can only route or filter the information agents produce. AEIC-MD addresses the prior question of whether they produce a verifiable, costly report in the first place. This ordering provides a principled way to retain prior topology assets without letting them displace the main contribution.

Limitations. The primary formal comparison uses one backend and three QA domains. The action menu and reward language are prompt-mediated implementations of costs and credits, rather than monetary transfers. The boundary centre points are development data, and the held-out-backend result prevents a broad generalization claim. Future work should pre-register backend-specific calibration, test mixed-model panels, and then study how an incentive-compatible reporting mechanism interacts with topology routing.

9 Conclusion

AEIC-MD treats structured AEIC reports as the observable interface of a multi-agent mechanism. In paired formal experiments, a public Static-AEIC rule substantially raises both revealed high-effort behavior and decision quality, whereas outcome-only reward does not improve quality. Reward–cost responses, controlled information conditions, seeds, and panel-size tests support the behavioral interpretation. At the same time, the negative GLM replication limits the claim to the tested backend and makes contract calibration across model families the central next step. The appropriate main story is therefore mechanism-mediated follower response—with topology as a subsequent information-control layer—not topology optimization alone.

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